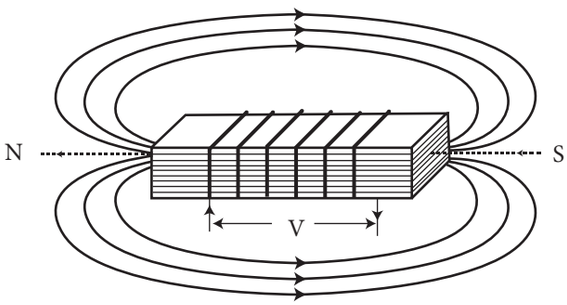
Electromagnetism

When a current is passed through a coil of wire, a magnetic field is setup around the coil. If soft iron bar is placed inside the coil of wire carrying current, the iron bar becomes magnetized. This process is known as electro magnetism.



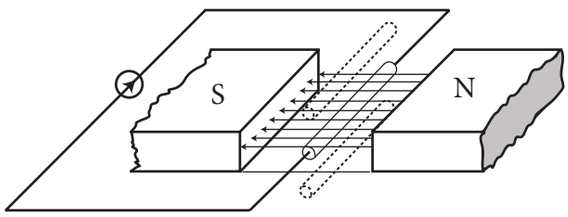
Magnetic Flux

Magnetic Flux

Magnetic flux is a group of lines of force crossing the space of a magnetic field. It is denoted by Ø. The unit of magnetic flux is weber in M.K.S system and Maxwell in C.G.S system.

Electro Magnetic Induction

The current is induced in a conductor when it is cut by a magnetic flux, it is known as Electro Magnetic induction. The current induced is known as Electro Motive Force (E.M.F). The e.m.f induced in the conductor depends upon the strength of the magnetic flux and the speed at which the conductor cuts the flux.



Faraday's Laws

10.4 FARADAY'S LAWS OF ELECTROMAGNETIC INDUCTION

First Law

Whenever any conductor is made to rotate in a magnetic field and hence to cut the magnetic lines of force or the flux, an electro motive force (e.m.f) will be induced in that conductor.

Second Law

The magnitude of the induced e.m.f is directly proportional to the rate of change of flux linked with the conductor.

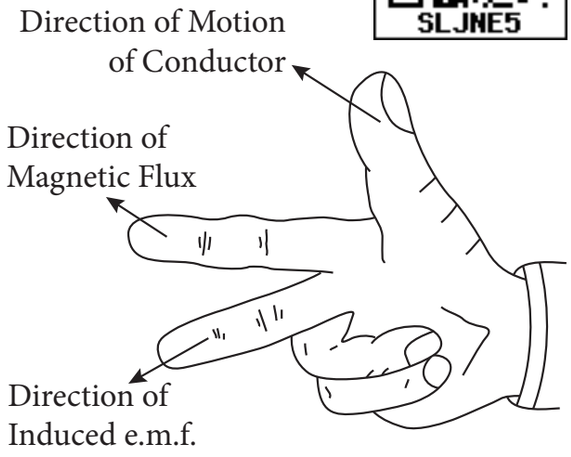
Direction of E.M.F

The relation between the directions of the motion of conductor, the induced e.m.f and the magnetic flux can be explained by Flemming's right hand and left hand rules.

10.5 FLEMMING RULES

Flemming's Right Hand Rule

SLJNE5



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